Created by Latif

A Data Modeling & Visualization Project using Supabase, dbt, and Superset.

SALES ANALYTICS DASHBOARD

Strategic Insights from Sales Performance and Discount Contribution.

July, 21 2025

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PROJECT OVERVIEW

This project explores a comprehensive sales dataset to uncover key performance drivers through data cleaning, modeling, and visualization. Using a modern data stack, including Supabase, dbt, and Superset (via Preset.io). I built an OLAP-based data model and interactive dashboard to monitor product performance, revenue trends, and discount impact.

The goal is to empower stakeholders with actionable insights to optimize sales strategy, evaluate promotion effectiveness, and enhance revenue consistency.

Documentation:

- Github Repo
- Dashboard

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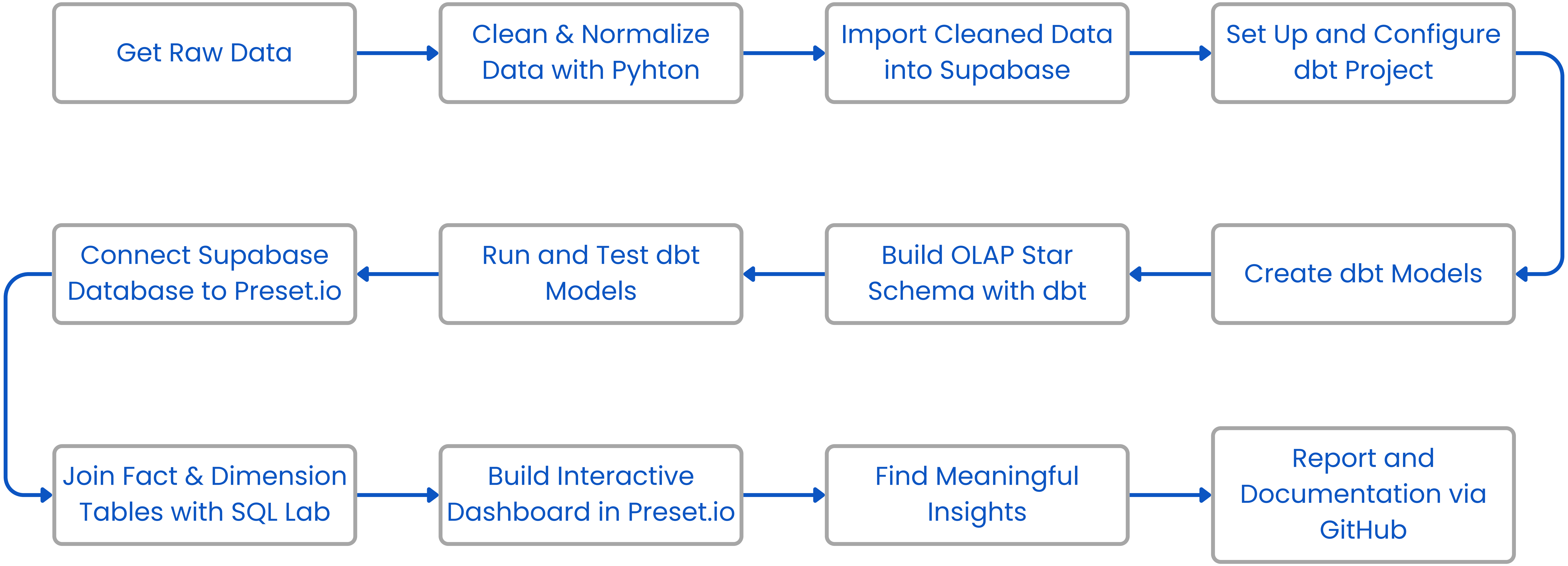
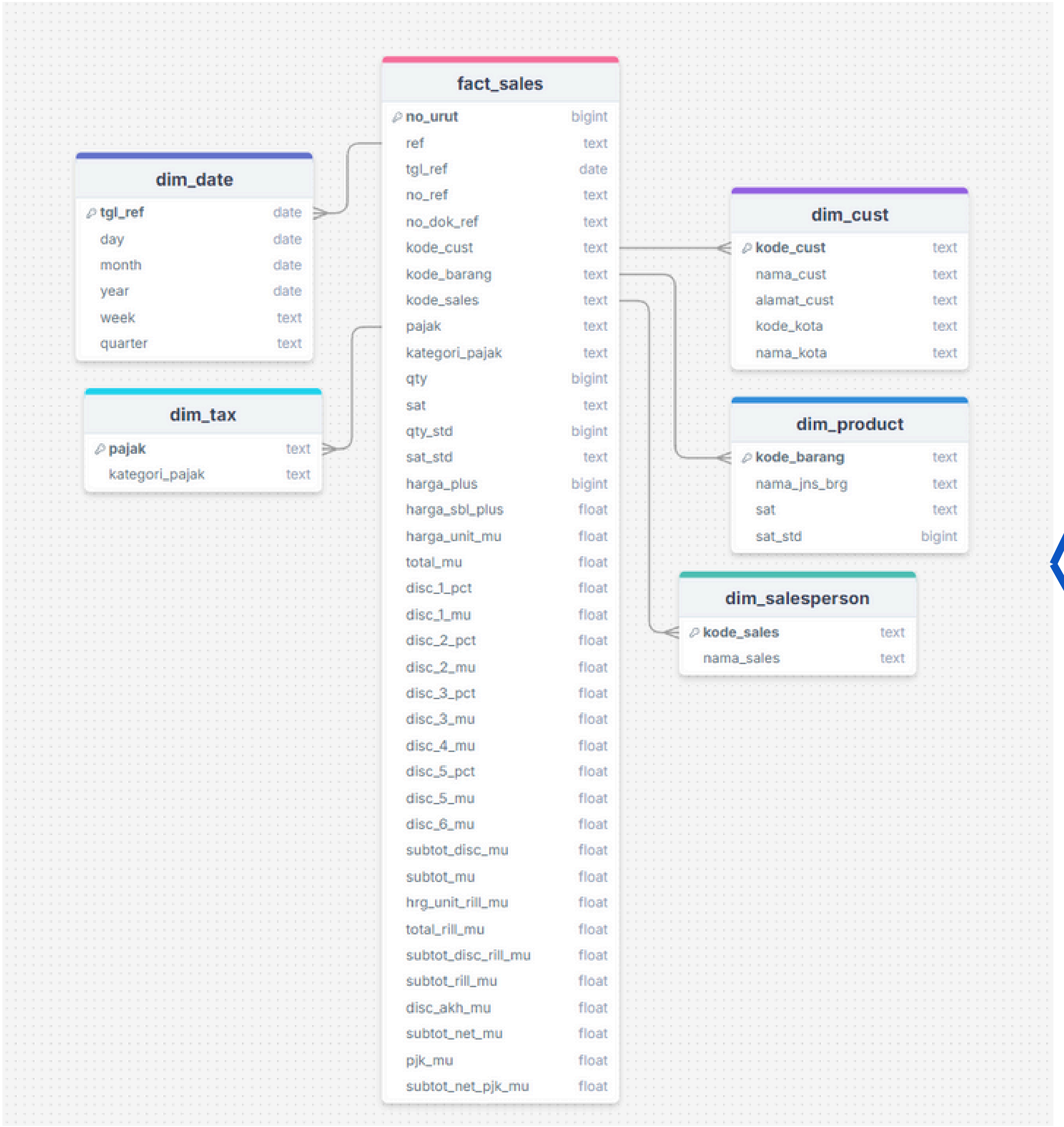
WORKFLOW

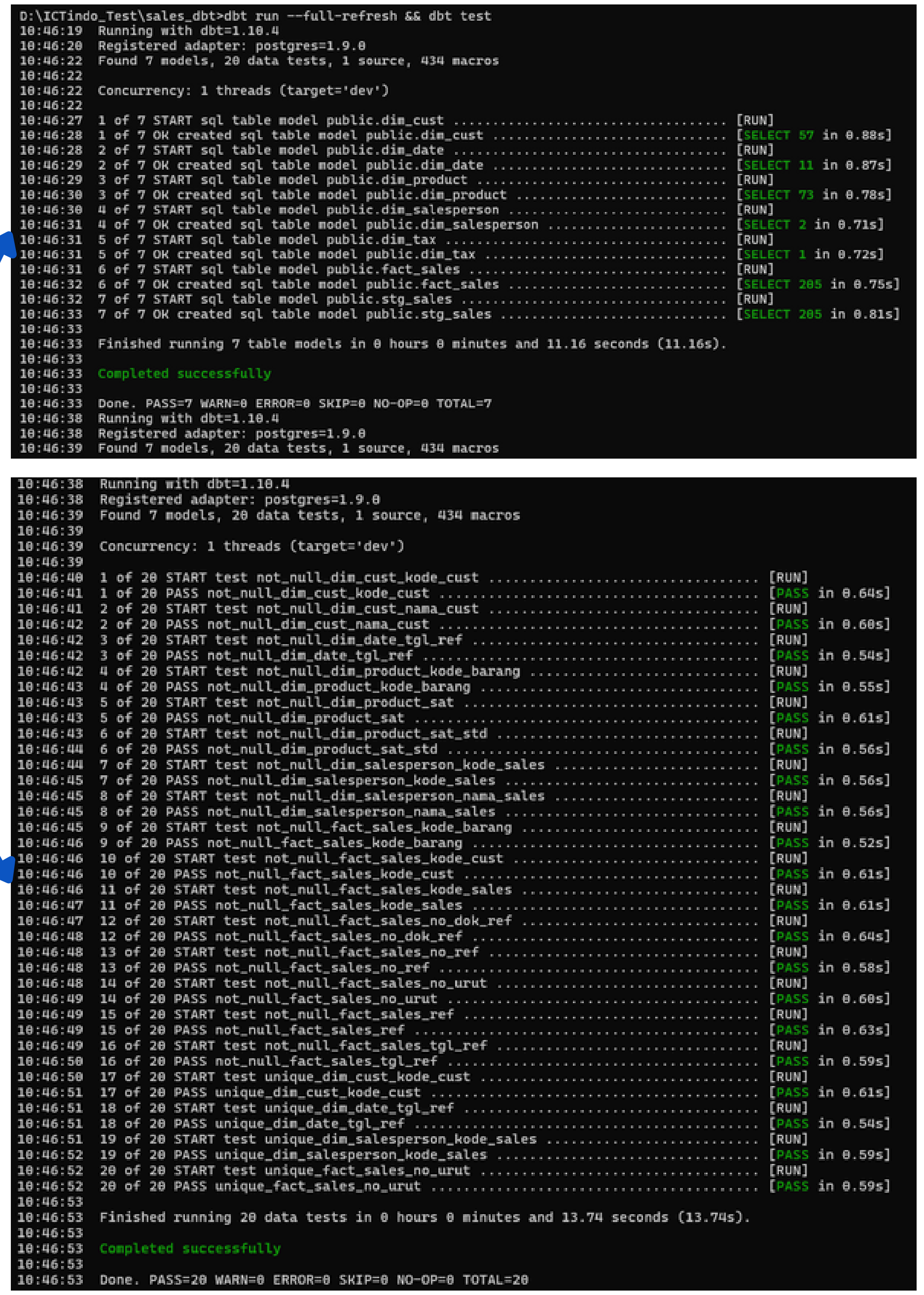


Tools used: Excel, Python, PostgreSQL, Supabase, dbt, Preset.io, GitHub

DATABASE SCHEMA



build dbt model with OLAP STAR Schema



running and testing dbt models

This database was designed using the **OLAP Star Schema** approach, with a central fact table **fact_sales** that records key sales metrics, surrounded by six dimension tables that provide descriptive context:

- **dim_cust**: customer information
- **dim_date**: date and time details
- **dim_product**: product details
- **dim_salesperson**: sales person information
- **dim_tax**: tax information

Using dbt, all data transformations are modular, documented, and version controlled ensuring a clean, consistent, and analysis-ready data pipeline. This structure enables efficient querying and seamless integration with BI tools.

PROBLEM STATEMENT & OBJECTIVES

Problem Statement:

The business currently holds raw sales data that lacks structure and analytical readiness. In its current form, the data cannot support effective reporting or decision-making processes. To drive actionable insights, the data needs to be cleaned, transformed, and organized into a scalable model that enables flexible exploration across time, product, customer, and sales channels.

Objectives:

1. Design an OLAP-style data model
Structure the raw sales dataset into a well-defined star schema consisting of a central fact table (fact_sales) and supporting dimension tables (dim_product, dim_cust, dim_date, dim_salesperson, and dim_tax) for efficient reporting and analysis.

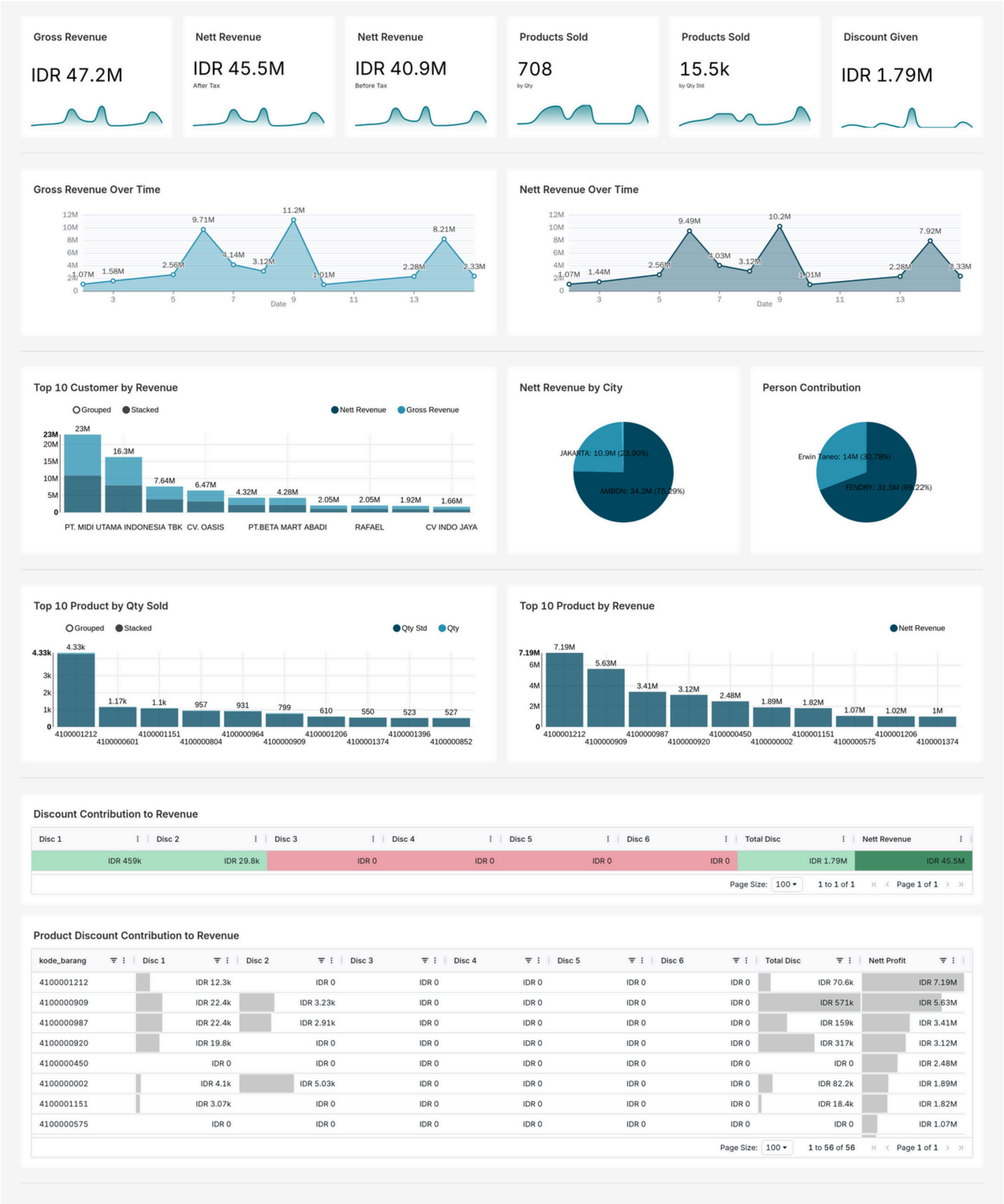
2. Build dbt models for data transformation
Use dbt to create modular, documented, and tested transformation models that convert raw data into reporting-ready tables.

3. Deploy data to a PostgreSQL database (Supabase)
Load both raw and transformed data into Supabase as the central storage for analysis and BI connectivity.

4. Develop an interactive dashboard using Superset (Preset.io)
Visualize insights with filters, KPIs, and charts that are clear, insightful, and actionable for business users.

5. Document and communicate
Provide clear documentation (README.md, schema.yml) to explain data structure, assumptions, and business logic behind the dashboard.

DASHBOARD RESULT



This dashboard presents a clear overview of sales performance, built from cleaned and transformed raw data using dbt in an OLAP star schema. Visualized with Preset (Superset), it enables stakeholders to track revenue trends, product performance, customer contributions, and the impact of discounts effectively.

Key Components of the Dashboard:

📈 KPI Summary

Displays Gross Revenue, Nett Revenue, Products Sold, and Discount Given.

📅 Revenue Trends

Line charts to track gross and net revenue over time for pattern and seasonality analysis.

👥 Customer & City Insights

Highlights top customers by revenue across cities.

🧑‍💼 Salesperson Contribution

Shows Sales contribution per salesperson.

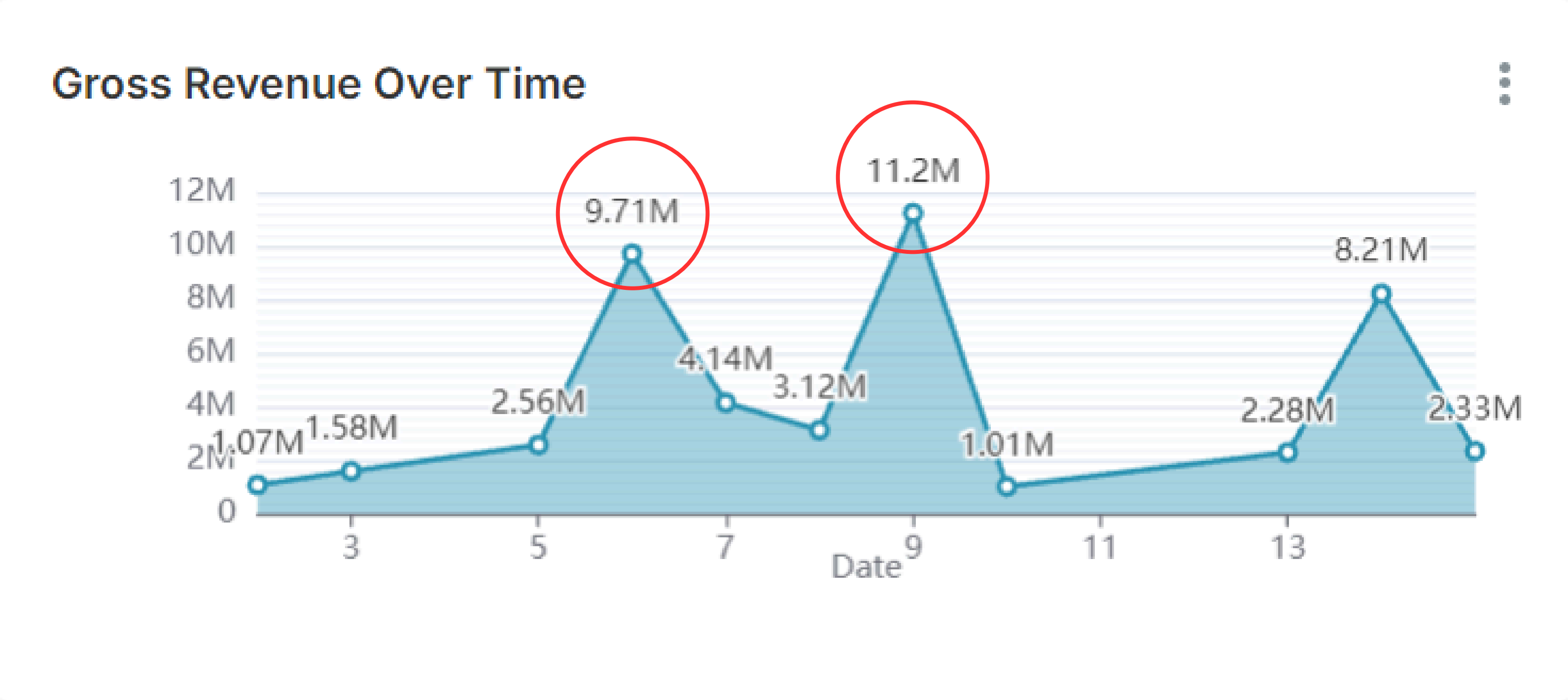
📦 Product Performance

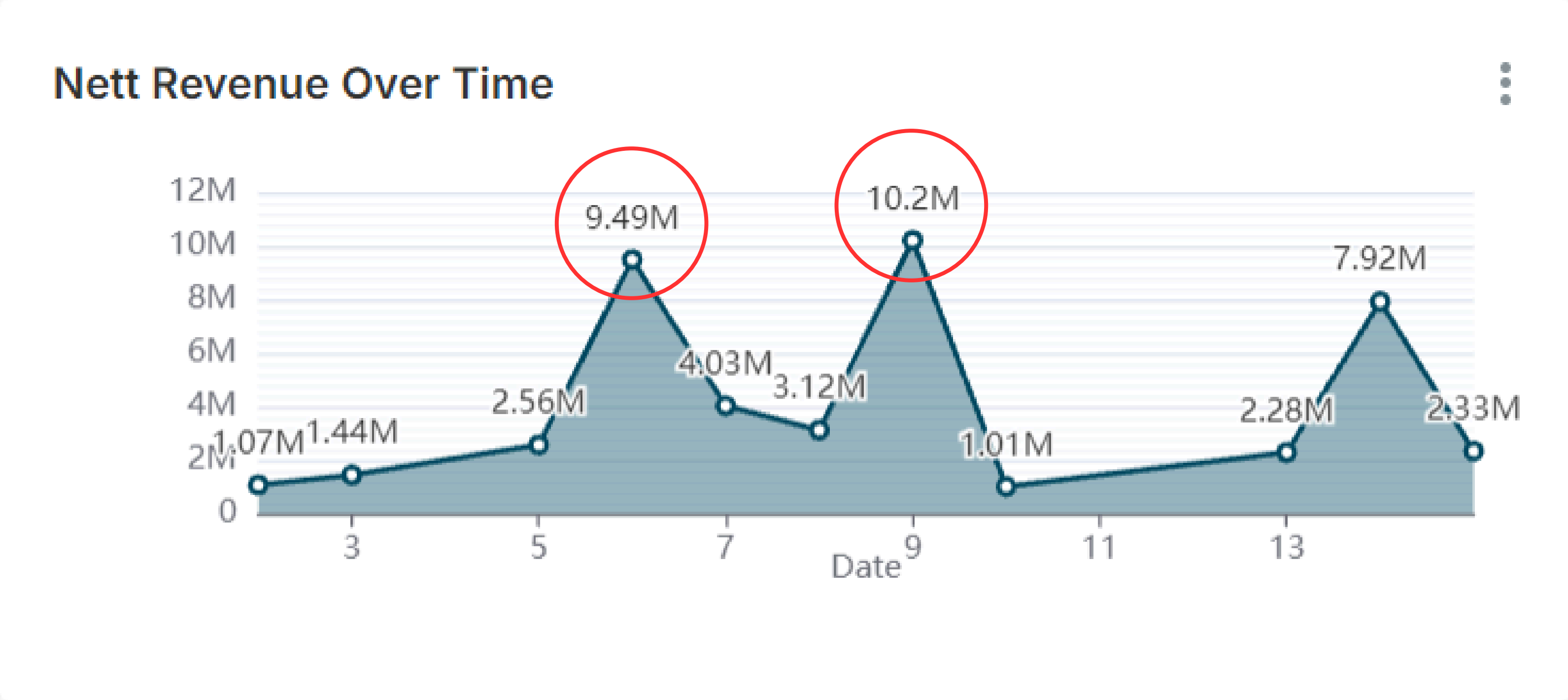
Lists top products by quantity sold and by revenue.

🎯 Discount Impact

Visualizes discount contributions to revenue and net profit at the product level.

KEY INSIGHTS & RECOMMENDATION | Revenue Trends





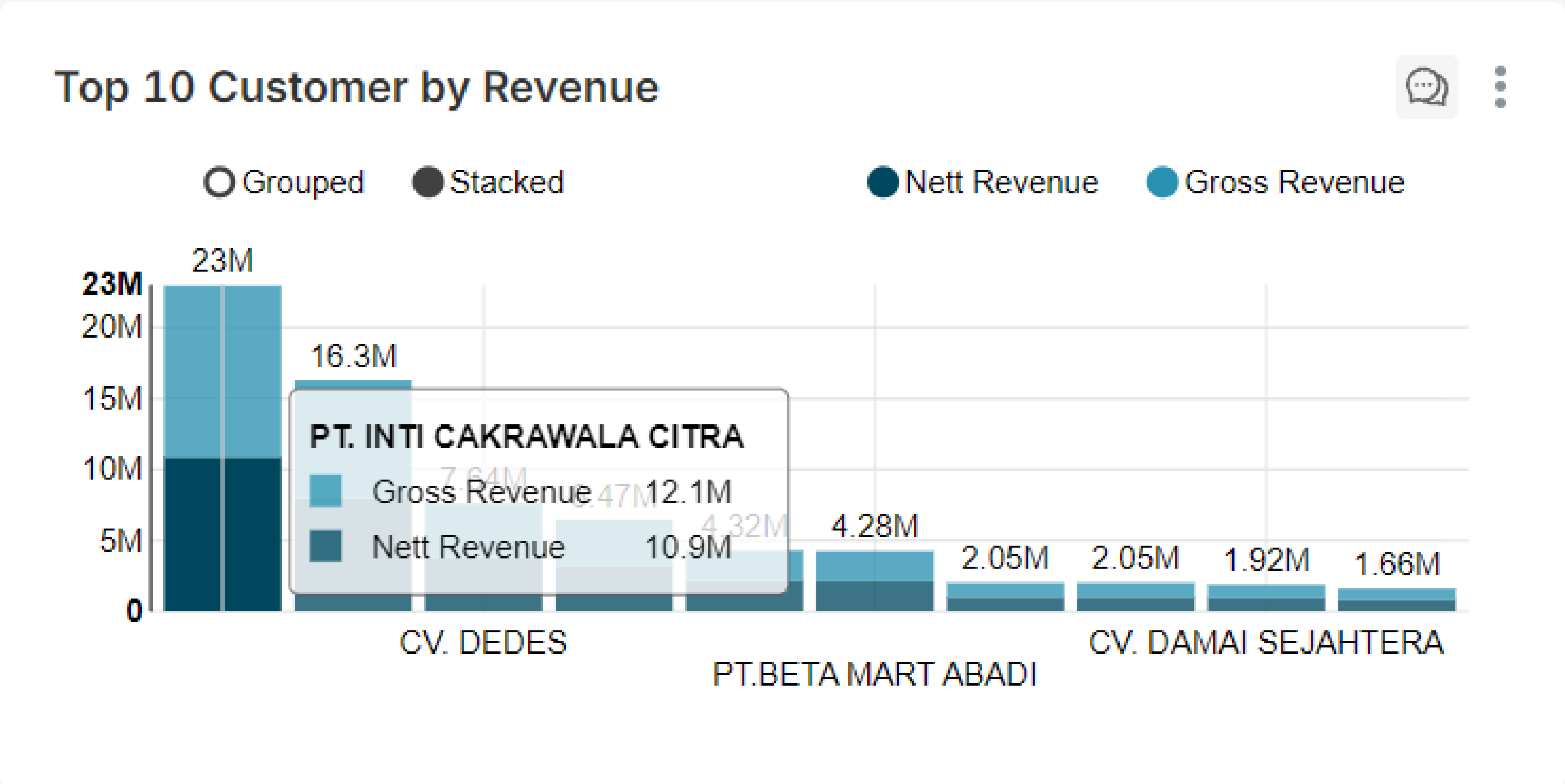
key Insights:

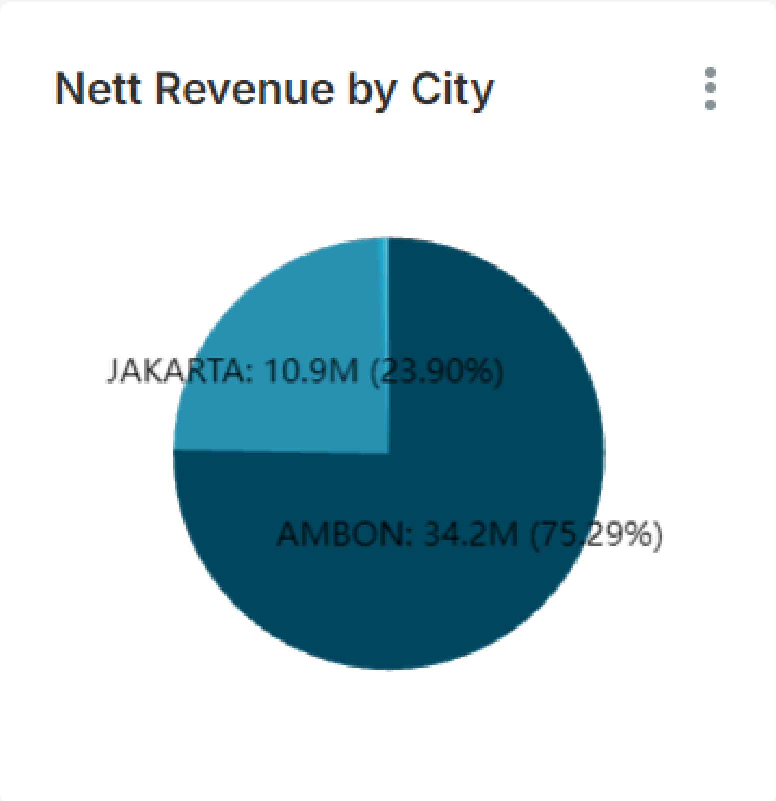
The sales trend reveals notable peaks in both gross and net revenue on May 6 and May 9, indicating high-performing days that significantly impact overall revenue. Meanwhile, several other dates reflect relatively low or stagnant sales activity. The gap between gross and net revenue appears proportionally stable, suggesting consistent tax application.

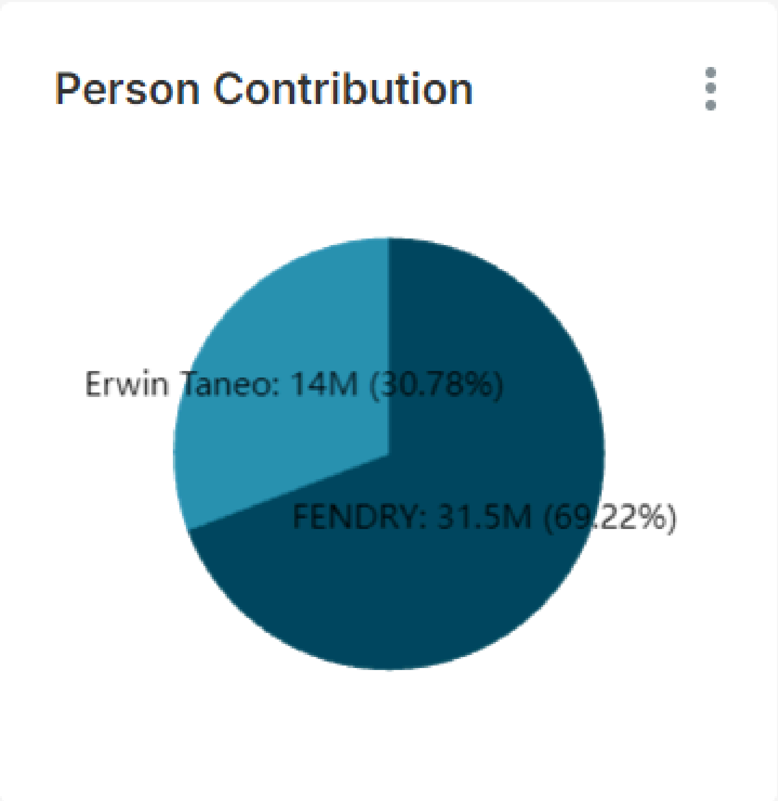
Recommendation:

Management should evaluate the factors contributing to the revenue spikes on May 6 and May 9, including campaign performance, product demand, or specific customer activity. It is also important to assess low-revenue days to identify any operational gaps or external disruptions. Increasing sales consistency across all days can help drive more stable and sustainable revenue growth.

KEY INSIGHTS & RECOMMENDATION | Customer and City Insight, Salesperson Contribution







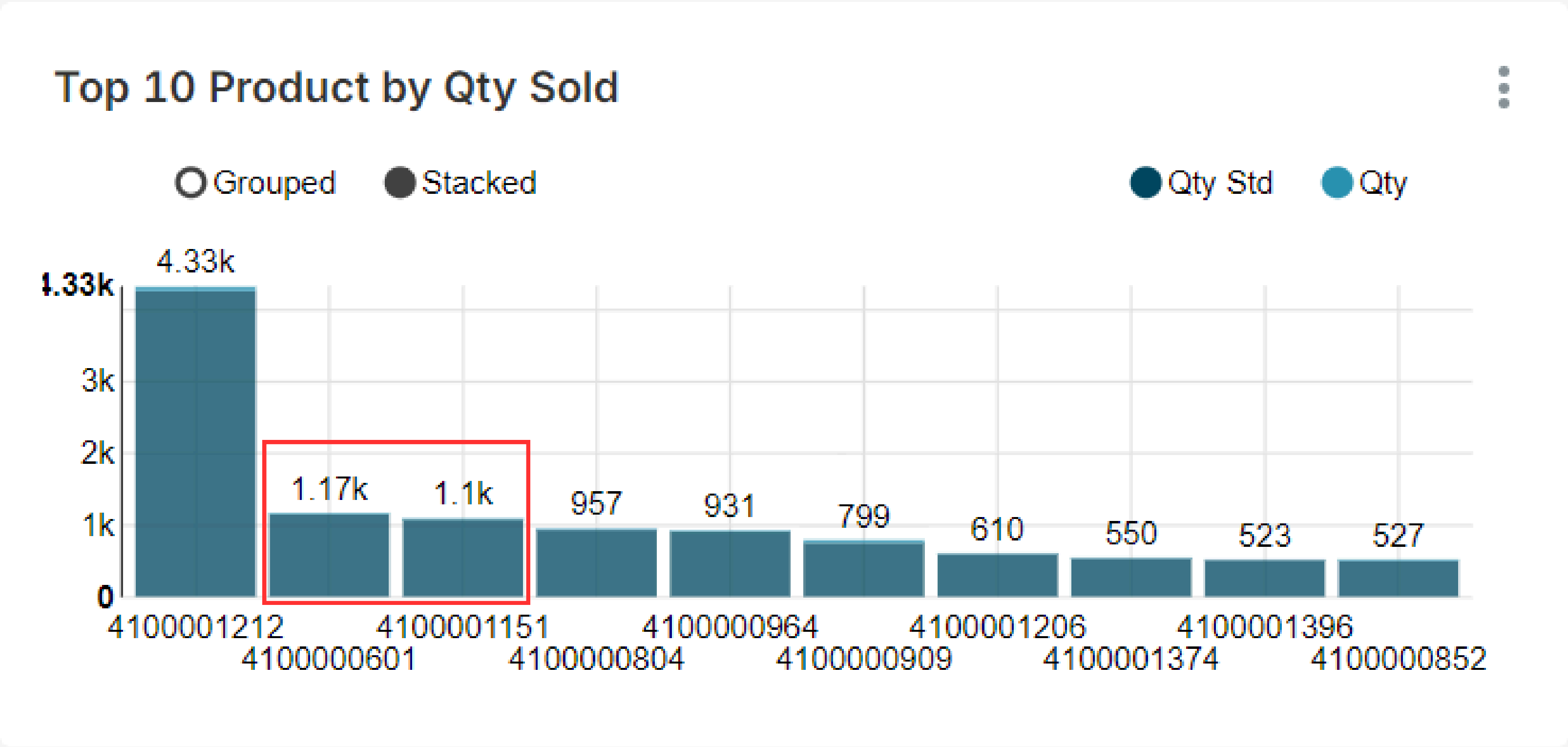
key Insights:

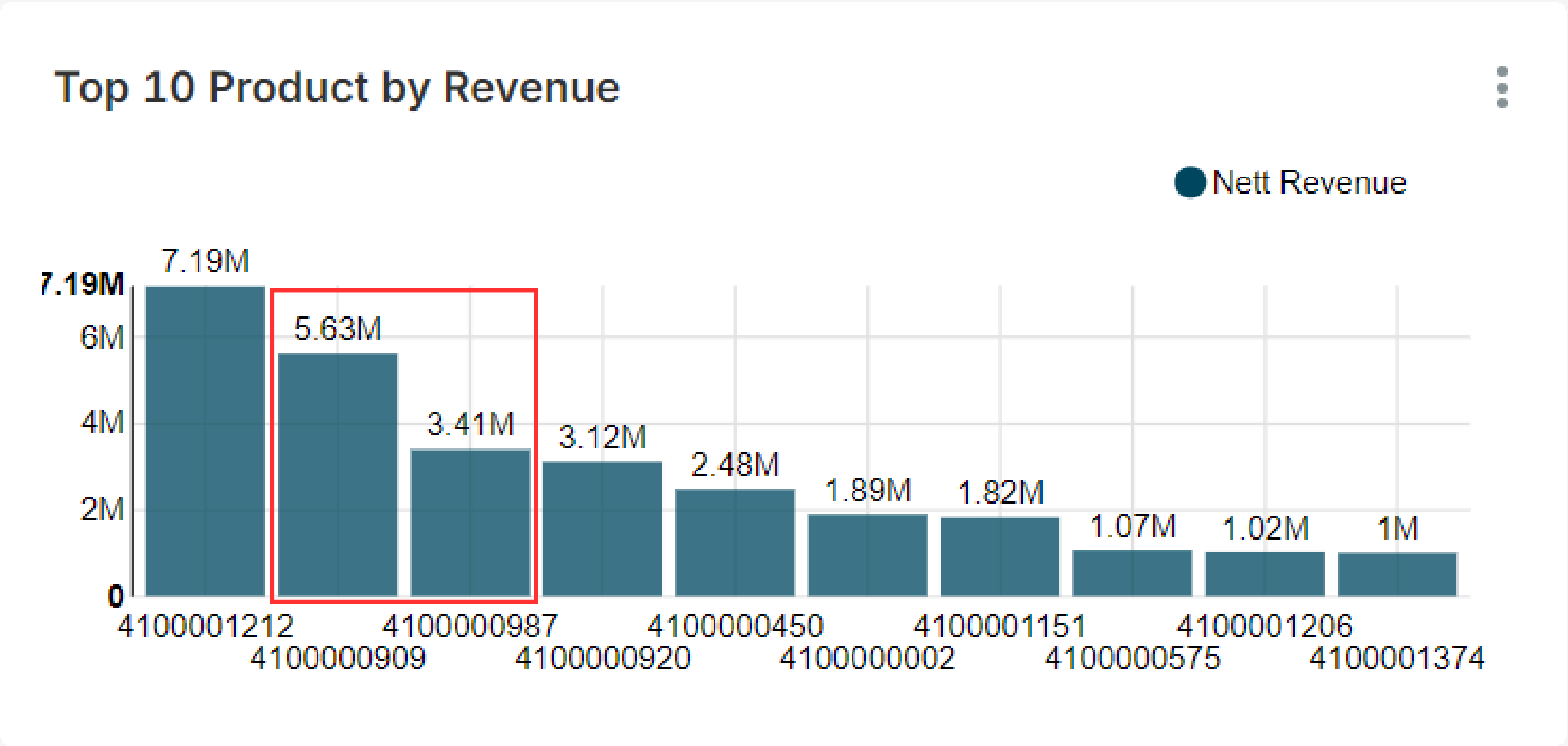
- PT. Inti Cakrawala Citra is the top contributors to gross and nett revenue, generating IDR 12.1M and IDR 10.9M.
- Ambon dominates city-level revenue performance with IDR 34.2M, far surpassing Jakarta (IDR 10.9M).
- Fendry contributes more than twice as much as Erwin Taneo, with IDR 31.5M in net revenue compared to IDR 14M, indicating a notable performance gap.

Recommendation:

- Prioritize retention strategies and personalized service for these key customers, while also identifying opportunities to increase revenue from mid-level customers.
- Strengthen distribution and sales efforts in Ambon to maintain momentum, while assessing growth potential in Jakarta and underperforming cities.
- Analyze the methods and strategies driving Fendry's success, and consider developing targeted training or best practices to enhance performance across the sales team.

KEY INSIGHTS & RECOMMENDATION | Product Performance





key Insights:

- Product 4100001212 leads in quantity sold with 4,33K units, far surpassing others like 4100000601 (1.17K) and 4100001151 (1,1K), indicating exceptionally high product movement.
- Product 4100001212 also generates the highest nett revenue at IDR 7.19M, indicating strong alignment between quantity sold and revenue impact.
- While 4100000601 and 4100001151 lead in quantity sold, their absence from the top revenue list indicates a lower value per unit than products like 4100000909 and 4100000987.

Recommendation:

- Focus on promoting products that perform well in both quantity and revenue, such as 4100001212, 4100000909 and 4100000987, to maximize overall sales impact.
- For products with high quantity but low revenue like 4100000601 and 4100001151, consider value adding strategies such as bundling, premium packaging, or selective price adjustments to increase revenue contribution.

KEY INSIGHTS & RECOMMENDATION | Discount Schema Impact

Discount Contribution to Revenue

Disc 1	Disc 2	Disc 3	Disc 4	Disc 5	Disc 6	Total Disc	Nett Revenue
IDR 459k	IDR 29.8k	IDR 0	IDR 0	IDR 0	IDR 0	IDR 1.79M	IDR 45.5M

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Product Discount Contribution to Revenue

kode_barang	Disc 1	Disc 2	Disc 3	Disc 4	Disc 5	Disc 6	Total Disc	Nett Profit
4100001212	IDR 12.3k	IDR 0	IDR 0	IDR 0	IDR 0	IDR 0	IDR 70.6k	IDR 7.19M
4100000909	IDR 22.4k	IDR 3.23k	IDR 0	IDR 0	IDR 0	IDR 0	IDR 571k	IDR 5.63M
4100000987	IDR 22.4k	IDR 2.91k	IDR 0	IDR 0	IDR 0	IDR 0	IDR 159k	IDR 3.41M
4100000920	IDR 19.8k	IDR 0	IDR 0	IDR 0	IDR 0	IDR 0	IDR 317k	IDR 3.12M
4100000450	IDR 0	IDR 0	IDR 0	IDR 0	IDR 0	IDR 0	IDR 0	IDR 2.48M
4100000002	IDR 4.1k	IDR 5.03k	IDR 0	IDR 0	IDR 0	IDR 0	IDR 82.2k	IDR 1.89M
4100001151	IDR 3.07k	IDR 0	IDR 0	IDR 0	IDR 0	IDR 0	IDR 18.4k	IDR 1.82M
4100000575	IDR 0	IDR 0	IDR 0	IDR 0	IDR 0	IDR 0	IDR 0	IDR 1.07M

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key Insights:

- Discount 1 dominates total discount value, followed by Discount 2. Discounts 3–6 show no usage, indicating low relevance or misalignment.
- Several top-selling products like 4100000450 and 4100000575 achieved high revenue without discounts, showing strong demand.

Recommendation:

- Review the design and relevance of Discount Schemes 3–6, as their lack of usage may indicate misalignment with current sales strategies or customer behavior.
- Leverage non-discounted top performers through optimized visibility and stock availability, as these products show potential for sustained revenue without relying on price cuts.

CONCLUSION

The analysis highlights the importance of balancing sales volume and profitability, as high-selling products do not always yield the highest revenue. It also reveals that only selected discount schemes significantly contribute to revenue, suggesting a need to focus future efforts on optimizing product mix and targeted promotions for better business impact.

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THANK YOU